

JEE Main - 9 | JEE - 2023 (Gen 1 & 2)

Date: 15/02/2022

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

SYLLABUS

Physics : Properties of Matter, Gaseous State & Thermodynamics
Chemistry: Stoichiometry II, IOC
Maths : Circles, Conic Sections

PART - I : PHYSICS

100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The density of a metal at temperature T_1 is ρ_1 and its density at temperature T_2 is ρ_2 . The volume thermal expansion coefficient of the metal is given by:

(A) $\gamma = \frac{3}{(T_2 - T_1)} \left(\frac{\rho_1}{\rho_2} - 1 \right)$ (B) $\gamma = \frac{3}{(T_2 - T_1)} \left(\frac{\rho_2}{\rho_1} - 1 \right)$
 (C) $\gamma = \frac{1}{(T_2 - T_1)} \left(\frac{\rho_1}{\rho_2} - 1 \right)$ (D) $\gamma = \frac{1}{(T_2 - T_1)} \left(\frac{\rho_2}{\rho_1} - 1 \right)$

2. Let the latent heat of fusion of ice be denoted by L , and let the specific heat of water be denoted by C . A mass m_1 of ice at 0°C is mixed with mass m_2 of liquid water at $T^\circ\text{C}$, and all the ice melts. Assuming that no heat is gained from or lost to the surroundings, the final temperature is:

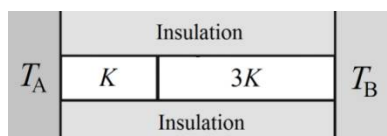
(A) $\frac{m_1 \left(\frac{L}{C} \right) - m_2 T}{(m_1 + m_2)}$ (B) $\frac{m_2 T - m_1 \left(\frac{L}{C} \right)}{(m_1 - m_2)}$ (C) $\frac{m_2 T - m_1 \left(\frac{L}{C} \right)}{(m_1 + m_2)}$ (D) $\frac{m_2 T + m_1 \left(\frac{L}{C} \right)}{(m_1 - m_2)}$

3. A body at temperature 80.0°C is placed in a room maintained at a constant temperature 20.0°C . The body takes time t_1 to cool from 80.0°C to 79.9°C , and it takes time t_2 to cool from 40.0°C to 39.9°C .

Assuming that the cooling process follows Newton's law of cooling, the ratio $\frac{t_1}{t_2}$ is close to:

(A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

4. Two cuboidal blocks made of different materials of thermal conductivity K and $3K$ are placed end to end between two heat reservoirs maintained at constant temperature T_A and T_B (such that $T_A > T_B$). The blocks have the same area of cross-section, but lengths L and $2L$ respectively. The temperature at the interface between the blocks is:



(A) $T = \frac{1}{7}(T_A + 6T_B)$ (B) $T = \frac{1}{7}(6T_A + T_B)$
 (C) $T = \frac{1}{5}(2T_A + 3T_B)$ (D) $T = \frac{1}{5}(3T_A + 2T_B)$

5. A simple pendulum consists of a metal wire of coefficient of linear thermal expansion α attached to a heavy bob. The time period of this pendulum on a cold winter day with temperature 5°C is T_C , and its time period on a hot summer day with temperature 35°C is T_H . The ratio $\frac{T_C}{T_H}$ is close to:

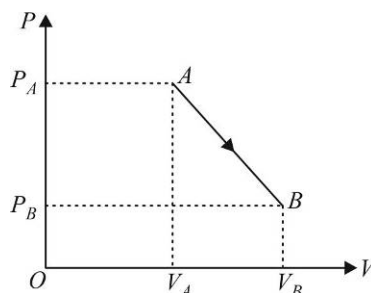
(A) $1 + 30\alpha$ (B) $1 - 30\alpha$ (C) $1 + 15\alpha$ (D) $1 - 15\alpha$

6. A wall is made up of two layers A and B. The thickness of the two layers is the same, but materials are different. The thermal conductivity of A is double than that of B. In thermal equilibrium the temperature difference between the two ends is 36°C . Then the difference of temperature at the two surfaces of A will be:
(A) 6°C **(B)** 12°C **(C)** 18°C **(D)** 24°C
7. A diatomic gas initially at pressure P_1 expands such that its pressure and volume follow the relation: $\frac{P}{V^3} = \text{constant}$. If volume of the gas increases from V_1 to V_2 during the process, the heat absorbed by the gas is:
(A) $\left(\frac{7}{4}\right)\left(\frac{P_1}{V_1^3}\right)(V_2^4 - V_1^4)$ **(B)** $\left(\frac{11}{4}\right)\left(\frac{P_1}{V_1^3}\right)(V_2^4 - V_1^4)$
(C) $\left(\frac{7}{4}\right)\left(\frac{P_1}{V_1^2}\right)(V_2^3 - V_1^3)$ **(D)** $\left(\frac{11}{4}\right)\left(\frac{P_1}{V_1^2}\right)(V_2^3 - V_1^3)$
8. A 5 cm thick ice block is there on the surface of water in a lake. The temperature of air is -10°C ; how much time it will take to double the thickness of the block: ($L = 80 \text{ cal/g}$, $K_{\text{ice}} = 0.004 \text{ erg/s-k}$, $d_{\text{ice}} = 0.92 \text{ g cm}^{-3}$)
(A) 1 hour **(B)** 191 hours **(C)** 19.1 hours **(D)** 1.91 hours
9. The specific heat at constant volume (in J/kgK) of Argon gas (molar mass = 40 g/mol) is close to: ($R = 8.31 \text{ J/mol K}$)
(A) 276 **(B)** 290 **(C)** 312 **(D)** 409
10. A body cools from 60°C to 50°C in 10 minutes. If the room temperature is 25°C and assuming Newton's law of cooling to hold good, the temperature of the body at the end of the next 10 minutes will be:
(A) 38.5°C **(B)** 40°C **(C)** 42.85°C **(D)** 45°C
11. One mole of an ideal gas expands at a constant temperature of 300 K from an initial volume of 10 litres to a final volume of 20 litres. The work done in expanding the gas is: ($R = 8.31 \text{ J/mol-K}$) ($\log_e 2 = 0.693$)
(A) 750 Joules **(B)** 1728 Joules **(C)** 1500 Joules **(D)** 3456 Joules
12. One mole of a monoatomic gas at 27°C is allowed to expand isothermally such that its final volume is 8 times its initial volume. The work done by the gas during the expansion is: ($\log_e 2 = 0.693$)
(A) 576 J **(B)** 1728 J **(C)** 5183 J **(D)** 466 J
13. One mole of a perfect gas in a cylinder fitted with a piston has a pressure P, volume V and temperature T. If the temperature is increased by 1 K keeping pressure constant, the increase in volume is:
(A) $\frac{2V}{273}$ **(B)** $\frac{V}{91}$ **(C)** $\frac{V}{273}$ **(D)** V

14. Let us assume that the sun and another star A are perfect spheres and perfect black bodies. The radius of star A is four times the radius of the sun and it radiates heat energy at exactly the same rate as the sun. If the surface temperature of the sun is 6000 K, the surface temperature (in Kelvin) of the star A is:

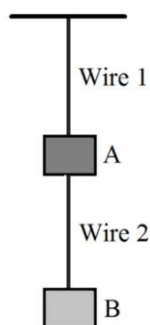
(A) 8484 (B) 4242 (C) 3000 (D) 2000

15. An ideal gas is taken from point A to the point B, as shown in the $P-V$ diagram. The work done in the process is:



(A) $(P_A - P_B)(V_B - V_A)$ (B) $\frac{1}{2}(P_B - P_A)(V_B + V_A)$
 (C) $\frac{1}{2}(P_B - P_A)(V_B - V_A)$ (D) $\frac{1}{2}(P_B + P_A)(V_B - V_A)$

16. Two blocks A and B of mass 4 kg and 2 kg respectively are suspended using two metallic wires as shown. The diameter of the wires are in the ratio $\frac{d_1}{d_2} = 1:2$, and their Young's moduli are in the ratio $\frac{Y_1}{Y_2} = 3:1$. The strain produced in them is in the ratio:



(A) 2 : 1 (B) 4 : 9 (C) 2 : 3 (D) 4 : 1

17. A monoatomic gas at initial pressure P_0 and initial temperature T_0 is allowed to expand isothermally until its volume doubles. Now, it is compressed adiabatically back to the initial volume. The final pressure and temperature are:

(A) $2^{2/3}P_0, 2^{2/3}T_0$ (B) $2^{2/3}P_0, 2^{5/3}T_0$ (C) $2^{5/3}P_0, 2^{2/3}T_0$ (D) $2^{5/3}P_0, 2^{5/3}T_0$

18. The specific heat of hydrogen gas at constant pressure is $C_P = 3.4 \times 10^3 \text{ cal/kg/C}$ and at constant volume is $C_V = 2.4 \times 10^3 \text{ cal/kg/C}$. If one kilogram hydrogen gas is heated from 10°C to 20°C at constant pressure, the external work done on the gas to maintain it at constant pressure is:

(A) 10^5 cal (B) 10^4 cal (C) 10^3 cal (D) $5 \times 10^3 \text{ cal}$

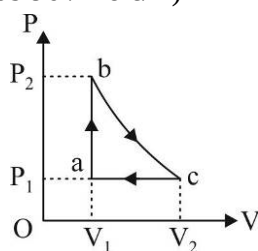
19. It takes 100 kJ heat to raise the temperature of a sample of an ideal monoatomic gas from 20°C to 120°C at constant pressure. The amount of heat needed (in kJ) to raise the temperature of the same sample of gas from 120°C to 220°C at constant volume is:
(A) 60 (B) 166.7 (C) 140 (D) 71.4
20. The pressure that has to be applied to the ends of a steel wire of length 10 cm to keep its length constant when its temperature is raised by 100°C is:
(For steel Young's modulus is $2 \times 10^{11} \text{ Nm}^{-2}$ and coefficient of thermal expansion is $1.1 \times 10^{-5} \text{ K}^{-1}$)
(A) $2.2 \times 10^8 \text{ Pa}$ (B) $2.2 \times 10^9 \text{ Pa}$ (C) $2.2 \times 10^7 \text{ Pa}$ (D) $2.2 \times 10^6 \text{ Pa}$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains **5 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. A 10 g ice cube at 0°C is added to an insulated vessel containing 150 g water at 80°C . Assuming that the heat capacity of the vessel is negligible, the final temperature (in $^\circ\text{C}$) of the mixture is _____.
(Latent heat of fusion of ice = 336 J/g , specific heat of water = 4.2 J/g)
22. A calorimeter of water equivalent 50 g contains 150 g water of 20°C . The amount of ice (in g) at -40°C that must be added to the calorimeter, so that the final temperature is 0°C with all the water frozen, is _____.
(Specific heat of water = $1\text{ cal/g}^\circ\text{C}$, specific heat of ice = $0.5\text{ cal/g}^\circ\text{C}$, latent heat of fusion of ice = 80 cal/g)
23. Carbon monoxide is carried around a closed cycle abc in which bc is an isothermal process as shown in the figure. The gas absorbs 70 KJ of heat as its temperature increases from 300 K to 1000 K in going from a to b. The work done (in KJ) on the gas (rounded off to the nearest two digit integer) during the process ca is: (Take the value of R as 8 J/mole/K)



24. A Carnot engine whose efficiency is 40%, receives heat at 500 K. If the efficiency is to be 50%, the source temperature (in Kelvin) for the same exhaust temperature is:
25. A gas with $\frac{C_p}{C_v} = \frac{3}{2}$ is heated at constant pressure. The ratio of the work done by the gas to the heat absorbed by the gas, $\frac{W}{\Delta Q}$ is equal to_____.

SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- How many moles of KMnO_4 are needed to oxidise a mixture of 1 mole of each FeSO_4 and FeC_2O_4 in acidic medium?
 (A) $\frac{4}{5}$ (B) $\frac{5}{4}$ (C) $\frac{3}{4}$ (D) $\frac{5}{3}$
- When the reaction: $\text{Cl}^- + \text{ClO}_3^- \longrightarrow \text{Cl}_2 + \text{H}_2\text{O}$ is balanced in acid solution, what is the ratio of Cl^- to ClO_3^- ?
 (A) $\frac{1}{1}$ (B) $\frac{2}{1}$ (C) $\frac{3}{1}$ (D) $\frac{5}{1}$
- What volume of $\text{O}_2(\text{g})$ measured at 1 atm and 273 K will be formed by action of 200 mL of 0.4 M KMnO_4 on hydrogen peroxide in acidic solution?
 (A) 4.48 litre (B) 2.24 litre (C) 8.96 litre (D) 2.12 litre
- 0.32 gram of N_2H_4 was oxidized by 100 mL, 0.1 M KIO_3 in conc. HCl where N_2H_4 is converted to N_2 . Possible conversion of KIO_3 is:
 (A) $\text{KIO}_3 \longrightarrow \text{I}_2$ (B) $\text{KIO}_3 \longrightarrow \text{ICl}$
 (C) $\text{KIO}_3 \longrightarrow \text{KIO}_4$ (D) $\text{KIO}_3 \longrightarrow \text{KI}$
- Consider the balanced chemical reaction :

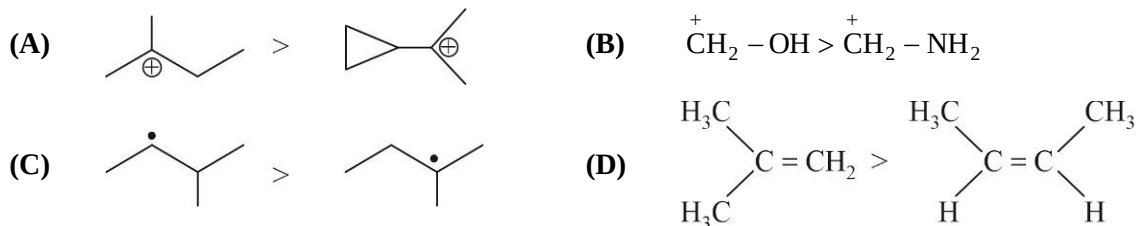
$$a\text{I}_2\text{O}_5 + b\text{BrF}_3 \longrightarrow c\text{IF}_5 + d\text{O}_2 + e\text{Br}_2$$
 Calculate the value of $(b + c + e) / (a)$.
 (A) 10 (B) 7 (C) 6 (D) 3
- 0.01 mole of KBrO_3 require 75 ml of standard $\text{Na}_2\text{S}_2\text{O}_3$ solution of molarity x M. Value of x is:
 [Reaction involved: (unbalanced) $\text{S}_2\text{O}_3^{2-} + \text{BrO}_3^- \rightarrow \text{SO}_4^{2-} + \text{Br}^-$]
 (A) 0.2 M (B) 0.1 M (C) 0.05 M (D) 0.01 M
- Which of the following reaction is redox ?
 (A) $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \longrightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$
 (B) $\text{CaC}_2 + 2\text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2 + \text{C}_2\text{H}_2$
 (C) $\text{Ca}(\text{OCl})\text{Cl} + \text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2 + \text{Cl}_2$
 (D) $\text{PCl}_5 + 4\text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$

8. Mass of KMnO_4 which is required to completely oxidise metal content of 3.92 gm. of the Mohr's salt ($\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$) is: [Molar mass :]

$$\text{KMnO}_4 = 158 \text{ g mol}^{-1}, \text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 = 284 \text{ g mol}^{-1}$$

- (A) 158 mg (B) 316 mg (C) 237 mg (D) 32 mg
9. A sample of a metal carbonate MCO_3 was neutralized by 10 ml of 0.1N – HCl and the resulting chloride gave 0.0517 g of phosphate, $\text{M}_3(\text{PO}_4)_2$. The equivalent weight of M is : [Atomic mass of P = 31]
- (A) 20.03 (B) 40.06 (C) 51.7 (D) 8.62

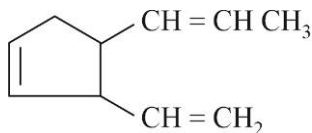
10. Which of the following order of stability is correct?



11. When methyl group is in axial position in methyl cyclohexane, the molecule has :

- (A) Meso isomer (B) Conformer
(C) Enantiomer (D) Diastereomer

12. Total number of stereoisomers of the given compound is :

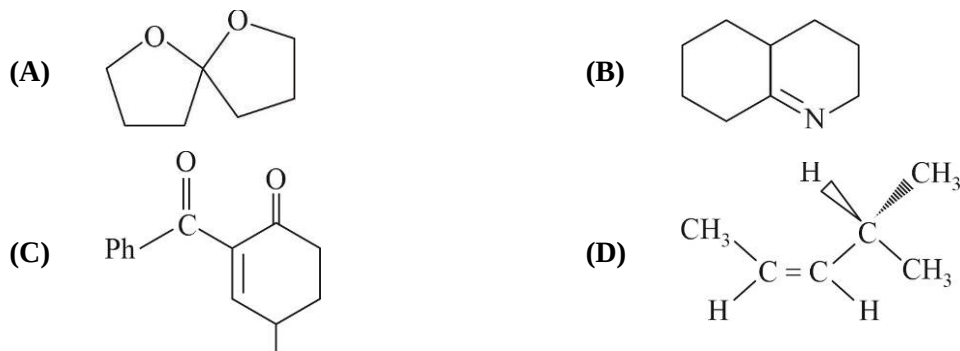


- (A) 6 (B) 8 (C) 16 (D) 4

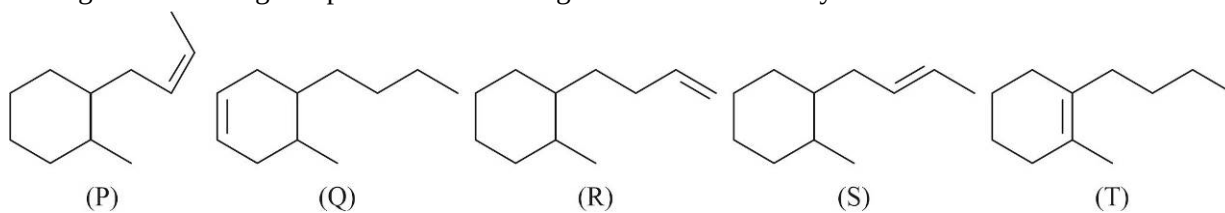
13. Total number of optical isomers of $\text{CH}_3 - \text{CHOH} - \text{CHOH} - \text{CHOH} - \text{CH}_3$ are :

- (A) 2 (B) 6 (C) 8 (D) 4

14. Which of the following molecule is not chiral ?

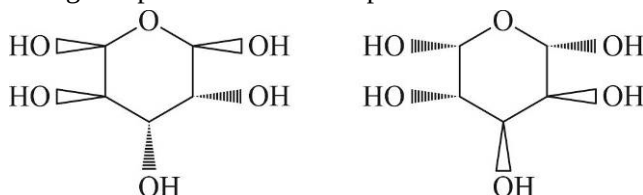


15. Arrange the following compounds in increasing order of their stability:



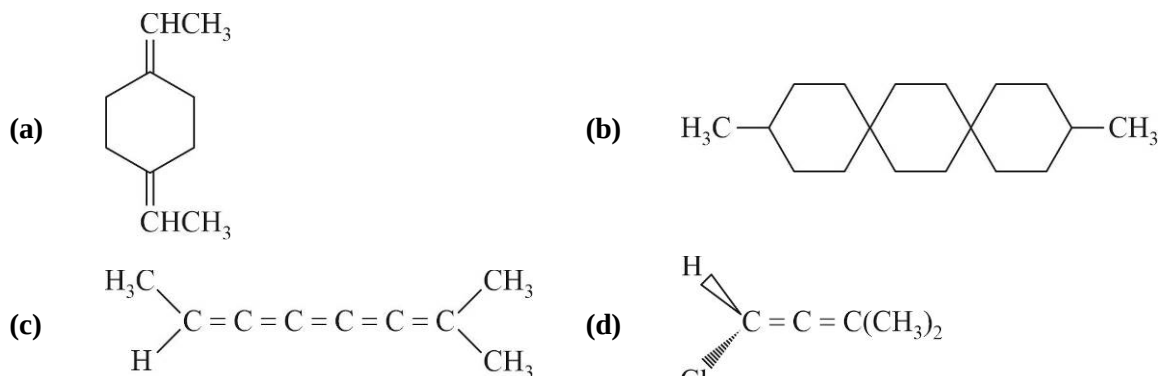
- (A) $R < Q < P < S < T$ (B) $T < S < P < Q < R$
 (C) $Q < P < R < T < S$ (D) $R < S < T < Q < P$

16. The following compounds differ in respect of :



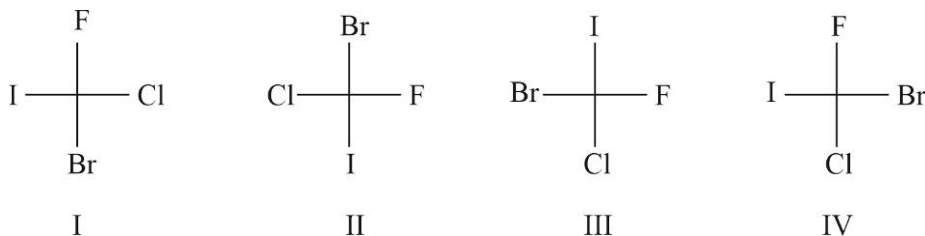
- (A) their chemical and physical properties
 (B) the direction in which they rotate plane of polarized light
 (C) their interaction with molecule
 (D) all are correct

17. Which of the following compounds are optically inactive but exhibit geometrical isomerism ?



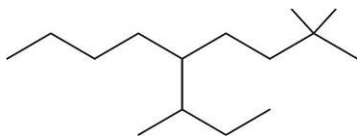
- (A) a, b, c only (B) a, b, c and d (C) a, b only (D) b, c only

18. Which of the following is a correct statement regarding the given molecules?



- (A) Compound I and II are diastereomers (B) Compound II and IV are enantiomers
 (C) Compound I and III are enantiomers (D) Compound I and IV are homomers

19. What is correct IUPAC name of the given compound ?



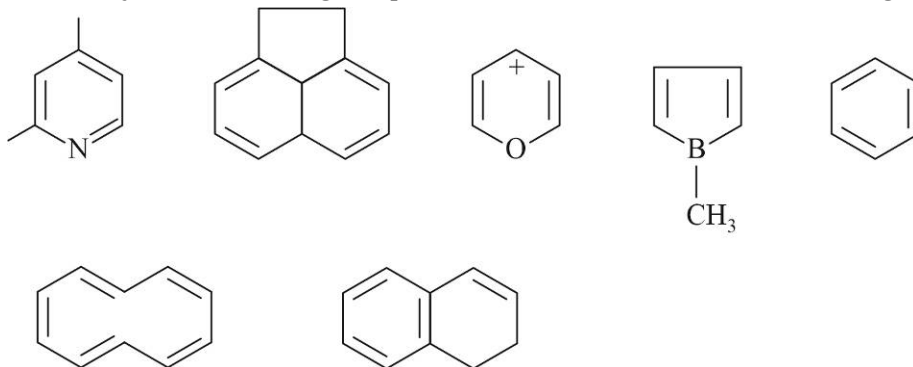
- (A) 4-Butyl-3, 7, 7-Trimethyloctane (B) 5-Butyl-2, 2, 6-Trimethyloctane
 (C) 6-methyl-5(3, 3-dimethylbutyl)octane (D) 2, 2-Dimethyl-5-(1-methylpropyl)nonane
20. In basic medium, Cl_2 undergoes disproportionation into Cl^- and ClO_x^- . If there is transfer of one mole of electron per mole of Cl_2 , the value of x is :
- (A) 3 (B) 1 (C) 2 (D) 4

SPACE FOR ROUGH WORK

SECTION-2

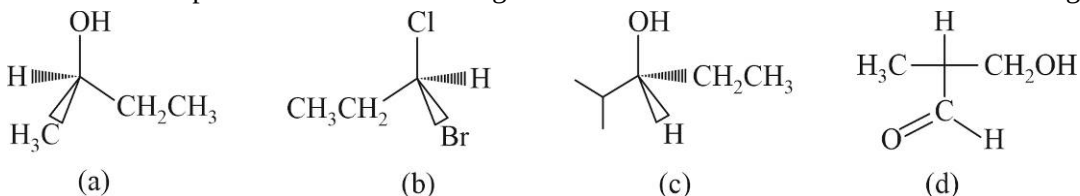
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21. How many of the following compounds have one/more aromatic resonating structures?



22. Number of possible isomers with molecular formula $C_4H_{10}O$ is:

23. Number of compounds from the following in which the chiral carbon atom have "S" configuration is:



24. A solution of 0.2 g of a compound containing Cu^{2+} and $C_2O_4^{2-}$ ions on titration with 0.02 M – $KMnO_4$ in the presence of H_2SO_4 consumes 22.6 ml of the oxidant. The resultant solution is neutralized with Na_2CO_3 , acidified with dilute acetic acid and treated with excess KI. It cause oxidation of iodide into iodine by Cu_{aq}^{2+} . The liberated iodine requires 11.3 ml of 0.05 M – $Na_2S_2O_3$ for complete reduction. If the molar ratio of Cu^{2+} to $C_2O_4^{2-}$ in the compound is 1 : x, the value of x is _____.
25. Sum of oxidation number of chromium atoms in one molecule each of the given compounds is:
 CrO_5 $Cr(C_6H_6)_2$ $[Cr(H_2O)_6]Cl_3$

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If the line $3x - 4y - k = 0$ touches the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ at (a, b) , then $k + a + b$ is equal to
 (A) 20 or -28 (B) 22 or -26
 (C) -30 or 24 (D) 28 or -20
- If the circle $C_1 : x^2 + y^2 = 16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to $3/4$, the co-ordinates of the centre of C_2 are
 (A) $\left(\frac{9}{5}, \frac{12}{5}\right)$ or $\left(-\frac{9}{5}, \frac{12}{5}\right)$ (B) $\left(-\frac{9}{5}, \frac{12}{5}\right)$ or $\left(-\frac{9}{5}, -\frac{12}{5}\right)$
 (C) $\left(\frac{9}{5}, -\frac{12}{5}\right)$ or $\left(-\frac{9}{5}, \frac{12}{5}\right)$ (D) $\left(\frac{9}{5}, \frac{12}{5}\right)$ or $\left(-\frac{9}{5}, -\frac{12}{5}\right)$
- The locus of the centres of the circles such that the point $(2, 3)$ is the mid point of the chord $5x + 2y = 16$ is
 (A) $2x - 5y + 11 = 0$ (B) $2x + 5y - 11 = 0$
 (C) $2x + 5y + 11 = 0$ (D) None of these
- The point of intersection of the lines $x - y + 1 = 0$ and $x + y + 5 = 0$ is P . A circle with centre at $(1, 0)$ passes through P . The tangent to the circle at P meets the x-axis at $(k, 0)$. The value of k is:
 (A) 2 (B) -3 (C) -2 (D) -4
- The equation of a circle which touches the line $x + y = 5$ at the point $(-2, 7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally is
 (A) $x^2 + y^2 + 7x - 11y + 38 = 0$ (B) $x^2 + y^2 + 7x + 11y + 38 = 0$
 (C) $x^2 + y^2 + 7x - 11y - 38 = 0$ (D) $x^2 + y^2 - 7x - 11y - 39 = 0$
- The area of the triangle formed by the tangents from the point $(4, 3)$ to the circle $x^2 + y^2 = 9$ and the line joining their points of contact is
 (A) 12 (B) 6 (C) 4 (D) $\frac{192}{25}$
- Let O be the centre of circle having radius 10 unit. Let P be fixed point inside the circle so that $OP = 6$. The number of chords passing through P and having integral length will be. (diameter is also a chord)
 (A) 4 (B) 8 (C) 7 (D) 9

8. The locus of the point of intersection of the normals at the ends of a focal chord of the parabola $y^2 = 4ax$ is
- (A) $y^2 = 4a(x - 3a)$ (B) $y^2 = 2a(x - 3a)$
 (C) $y^2 = a(x - 3a)$ (D) $y^2 = 16a(x - 3a)$
9. The mirror image of the directrix of the parabola $y^2 = 4(x + 1)$ in the line mirror $x + 2y = 3$ is:-
- (A) $x = -2$ (B) $4y + 3x = 16$
 (C) $3x - 4y + 16 = 0$ (D) None of these
10. Tangents are drawn to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ at ends of latus rectum. The area of quadrilateral so formed is:
- (A) 27 (B) $\frac{27}{2}$ (C) $\frac{27}{4}$ (D) $\frac{27}{55}$
11. The locus of the feet of the perpendicular drawn from the centre to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ on any tangent to it is
- (A) $(x^2 + y^2)^2 = 9x^2 + 4y^2$ (B) $(x^2 + y^2)^2 = 4x^2 + 9y^2$
 (C) $(x^2 + y^2)^2 = 3x^2 + 2y^2$ (D) $(x^2 + y^2)^2 = 2x^2 + 3y^2$
12. In an ellipse the distance between the foci is 8 and the distance between the directrices is 25, then the ratio of the length of major and minor axis is
- (A) $\frac{5}{\sqrt{17}}$ (B) $\frac{3}{\sqrt{17}}$ (C) $\frac{4}{\sqrt{17}}$ (D) $\frac{6}{\sqrt{17}}$
13. The maximum distance of perpendicular from the centre of the ellipse $4x^2 + 9y^2 = 36$ on the chord of contact of mutually perpendicular tangents of the ellipse is
- (A) $6/\sqrt{13}$ (B) $5/\sqrt{13}$ (C) $9/\sqrt{13}$ (D) None of these
14. The radius of the largest circle inscribed in the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$ and having centre at $(1, 0)$ is
- (A) $\frac{11}{3}$ (B) $\sqrt{\frac{11}{3}}$ (C) $\frac{7}{3}$ (D) $\sqrt{\frac{7}{3}}$
15. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices at A, M and the origin O is
- (A) $\frac{31}{10}$ (B) $\frac{29}{10}$ (C) $\frac{21}{10}$ (D) $\frac{27}{10}$

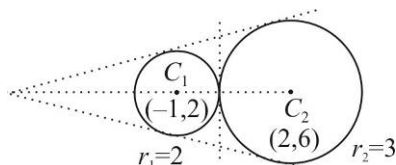
16. The co-ordinates of a point on the hyperbola $\frac{x^2}{24} - \frac{y^2}{18} = 1$, which is nearest to the line $3x + 2y + 1 = 0$ are
 (A) $(6, 3)$ (B) $(-6, -3)$ (C) $(6, -3)$ (D) $(-6, 3)$
17. Shortest distance between the curves $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{x^2}{4} + \frac{y^2}{4} = 1$
 (A) 1 (B) 2 (C) 3 (D) 4
18. A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$ is:
 (A) $y\sqrt{7} = \sqrt{2}x + 15$ (B) $y\sqrt{7} = 3\sqrt{2}x + 15$
 (C) $y = 3\sqrt{2}x + 15$ (D) $y\sqrt{7} = 3x + 15$
19. If the line $y = mx + \sqrt{a^2m^2 - b^2}$, $m = \frac{1}{2}$ touches the hyperbola $\frac{x^2}{16} - \frac{y^2}{3} = 1$ at the point $(4\sec\theta, \sqrt{3}\tan\theta)$, then θ is:
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{6}$
20. The number of possible tangents can be drawn to the curve $4x^2 - 9y^2 = 36$, which are perpendicular to the straight line $5x + 2y = 10$ is
 (A) 0 (B) 1 (C) 2 (D) 4

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains **5 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. The number of straight lines in $x-y$ plane, whose distance from the point $(-1, 2)$ is 2 unit and from $(2, 6)$ is 3 unit:



22. If L is the length of perpendicular drawn from the origin to any normal of the ellipse, $\frac{x^2}{25} + \frac{y^2}{16} = 1$ then maximum value of L is:
23. A circle has the same centre as an ellipse & passes through the foci F_1 & F_2 of the ellipse, such that the two curves intersect in 4 points. Let ' P ' be any of their point of intersection. If the major axis of the ellipse is 17 & the area of the triangle PF_1F_2 is 30, then the distance between the foci is:
24. If $y = m_1x - 3$, $y = m_2x + 1$ and $y = x + 4$ are three normals drawn from a fixed point to $y^2 = 4x$. Find $|4m_1 - 6m_2|$.
25. The number of points with integral co-ordinates that lie in the interior of the region common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 4x$ is:

SPACE FOR ROUGH WORK

End of JEE Main – 9 | JEE - 2023 (Gen 1 & 2)